Class 12 Mathematics

Chapter 2: Inverse Trigonometric Functions

School Worksheet: step-wise solutions

20 questions **43** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

The principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is:

(A) $-\frac{\pi}{6}$ **(B)** $\frac{7\pi}{6}$

(C) $\frac{5\pi}{6}$ **(D)** $\frac{\pi}{6}$

Formula / idea: Principal value branch of $\sin^{-1}$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and $\sin^{-1}(-x) = -\sin^{-1} x$.

1 Use $\sin^{-1}(-x) = -\sin^{-1} x$.

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\sin^{-1}\left(\frac{1}{2}\right)$$

2 Since $\sin\frac{\pi}{6} = \frac{1}{2}$ and $\frac{\pi}{6}$ lies in the principal branch: ✓ 1

$$-\sin^{-1}\left(\frac{1}{2}\right) = -\frac{\pi}{6}$$

$-\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

3 Hence the correct option is (A).

Answer **(A)** $-\frac{\pi}{6}$

Watch out: Choosing $\frac{7\pi}{6}$: its sine is also $-\frac{1}{2}$, but it is outside $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

2 MCQ — 1 mark

The value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ is:

(A) $\frac{7\pi}{6}$ **(B)** $\frac{5\pi}{6}$

(C) $\frac{\pi}{6}$ **(D)** $-\frac{\pi}{6}$

Formula / idea: $\cos^{-1}(\cos x) = x$ only when $x \in [0, \pi]$.

1 Check the angle: $\frac{7\pi}{6} > \pi$, so it is not in $[0, \pi]$.

$$\cos^{-1}\left(\cos\frac{7\pi}{6}\right) \neq \frac{7\pi}{6}$$

2 Rewrite $\cos\frac{7\pi}{6}$ using $\cos(2\pi-\theta)=\cos\theta$.

$$\cos\frac{7\pi}{6}=\cos\left(2\pi-\frac{7\pi}{6}\right)=\cos\frac{5\pi}{6}$$

3 Now $\frac{5\pi}{6}\in[0,\pi]$, so: ✓ 1

$$\cos^{-1}\left(\cos\frac{5\pi}{6}\right)=\frac{5\pi}{6}$$

4 Hence the correct option is (B).

Answer **(B)** $\frac{5\pi}{6}$

Watch out: Writing $\frac{7\pi}{6}$ directly: it lies outside the range $[0,\pi]$ of $\cos^{-1}$.

3 MCQ — 1 mark

The domain of the function $y=\sin^{-1}(2x)$ is:

(A) $[-1,1]$ **(B)** $[-\frac{1}{2},\frac{1}{2}]$

(C) $[-2,2]$ **(D)** $[0,1]$

Formula / idea: $\sin^{-1}t$ is defined only for $-1\le t\le 1$.

1 The input of $\sin^{-1}$ must lie in $[-1,1]$.

$$-1\le 2x\le 1$$

2 Divide each part by 2. ✓ 1

$$-\frac{1}{2}\le x\le\frac{1}{2}$$

3 Domain $=\left[-\frac{1}{2},\frac{1}{2}\right]$. Hence option (B).

Answer **(B)** $\left[-\frac{1}{2},\frac{1}{2}\right]$

Watch out: Answering $[-1,1]$: that is the condition on $2x$, not on x.

4 MCQ — 1 mark

The range of the principal value branch of $\sec^{-1}x$ is:

(A) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ **(B)** $(0,\pi)$

(C) $[0,\pi]-\left\{\frac{\pi}{2}\right\}$ **(D)** $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{0\}$

Formula / idea: $\sec^{-1}:\mathbb{R}-(-1,1)\to[0,\pi]-\left\{\frac{\pi}{2}\right\}$

1 Since $\sec\theta=\frac{1}{\cos\theta}$, $\sec^{-1}$ uses the same branch as $\cos^{-1}$, i.e. $[0,\pi]$.

2 Remove $\frac{\pi}{2}$, because $\cos\frac{\pi}{2}=0$ and $\sec\frac{\pi}{2}$ is not defined. ✓ 1
$\text{Range of } \sec^{-1}x=[0,\pi]-\left\{\frac{\pi}{2}\right\}$

3 Hence the correct option is (C).

Answer **(C)** $[0,\pi]-\left\{\frac{\pi}{2}\right\}$

Watch out: Picking (D): that is the range of $\csc^{-1}x$, not $\sec^{-1}x$.

5 MCQ 1 mark

The principal value of $\sec^{-1}\left(\sqrt{2}\right)+2\csc^{-1}\left(-\sqrt{2}\right)$ is:

(A) $-\frac{\pi}{2}$ **(B)** $-\frac{\pi}{4}$

(C) $\frac{\pi}{4}$ **(D)** $\frac{\pi}{2}$

Formula / idea: Ranges: $\sec^{-1}$ is $[0,\pi]-\left\{\frac{\pi}{2}\right\}$; $\csc^{-1}$ is $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{0\}$.

1 $\sec\frac{\pi}{4}=\sqrt{2}$ and $\frac{\pi}{4}$ is in the range of $\sec^{-1}$.
$\sec^{-1}\left(\sqrt{2}\right)=\frac{\pi}{4}$

2 $\csc\left(-\frac{\pi}{4}\right)=-\sqrt{2}$ and $-\frac{\pi}{4}$ is in the range of $\csc^{-1}$.
$\csc^{-1}\left(-\sqrt{2}\right)=-\frac{\pi}{4}$

3 Add the values. ✓ 1
$\frac{\pi}{4}+2\left(-\frac{\pi}{4}\right)=\frac{\pi}{4}-\frac{\pi}{2}=-\frac{\pi}{4}$

4 Hence the correct option is (B).

Answer **(B)** $-\frac{\pi}{4}$

Watch out: Taking $\csc^{-1}\left(-\sqrt{2}\right)=\frac{3\pi}{4}$ or $\frac{5\pi}{4}$: these are outside the principal branch.

6 **Assertion-Reason** 1 mark

Assertion (A): The domain of $y = \cos^{-1} x$ is $[-1, 1]$. **Reason (R):** The range of the principal value branch of $y = \cos^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $\cos^{-1} : [-1, 1] \to [0, \pi]$

1	Assertion: $\cos x$ takes values only in $[-1, 1]$, so domain of $\cos^{-1} x$ is $[-1, 1]$; **A is true.**	
2	Reason: range of $\cos^{-1} x$ is $[0, \pi]$, and $\frac{\pi}{2}$ is included; **R is false.** $\cos^{-1}(0) = \frac{\pi}{2}$	✓ 1
3	R is false, so it cannot explain A.	
4	Hence the correct option is (C).	

Answer **(C) A is true, but R is false**

Watch out: $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ is the range of $\sec^{-1} x$; do not mix it with $\cos^{-1} x$.

7 **Short answer** 2 marks

Find the value of $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$.

Formula / idea: $\sin^{-1}(\sin x) = x$ only when $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

1	Check the angle: $\frac{3\pi}{5} > \frac{\pi}{2}$, so it is not in the principal branch. $\frac{3\pi}{5} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	✓ ½
2	Use $\sin(\pi - \theta) = \sin\theta$. $\sin\frac{3\pi}{5} = \sin\left(\pi - \frac{3\pi}{5}\right) = \sin\frac{2\pi}{5}$	✓ ½
3	Check the new angle. $\frac{2\pi}{5} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	✓ ½

4 Therefore:

$$\sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \sin^{-1}\left(\sin\frac{2\pi}{5}\right) = \frac{2\pi}{5}$$

✓ ½

Answer $\frac{2\pi}{5}$

Watch out: Writing the answer as $\frac{3\pi}{5}$ without checking the principal branch.

Level 2

16 marks

8 **Short answer** 2 marks

Evaluate: $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) + \tan^{-1}\left(\tan\frac{2\pi}{3}\right)$.

Formula / idea: Ranges: $\tan^{-1}$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; $\cot^{-1}$ is $(0, \pi)$.

1 Find the first two terms.

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$

$$\cot^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{3}$$

$\cot\frac{\pi}{3} = \frac{1}{\sqrt{3}}$ and $\frac{\pi}{3} \in (0, \pi)$

✓ ½

2 Third term: $\sin\left(-\frac{\pi}{2}\right) = -1$.

$$\tan^{-1}(-1) = -\tan^{-1}(1) = -\frac{\pi}{4}$$

✓ ½

3 Fourth term: $\frac{2\pi}{3}$ is outside $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; use $\tan(\theta - \pi) = \tan\theta$.

$$\tan^{-1}\left(\tan\frac{2\pi}{3}\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3}$$

✓ ½

4 Add all four values.

$$-\frac{\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4} - \frac{\pi}{3}$$

$$= \frac{-2\pi - 3\pi}{12} = -\frac{5\pi}{12}$$

✓ ½

Answer $-\frac{5\pi}{12}$

Watch out: Writing $\tan^{-1}\left(\tan\frac{2\pi}{3}\right) = \frac{2\pi}{3}$; the principal branch of $\tan^{-1}$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

9 **Short answer** 2 marks

Find the domain of $f(x) = \sin^{-1}\left(x^2 - 4\right)$.

Formula / idea: $\sin^{-1} t$ is defined only for $-1 \le t \le 1$.

1	The input of $\sin^{-1}$ must lie in $[-1, 1]$. $-1 \le x^2 - 4 \le 1$	✓ ½
2	Add 4 to each part. $3 \le x^2 \le 5$	✓ ½
3	Take square roots; $\sqrt{x^2} = \lvert x \rvert$. $\sqrt{3} \le \lvert x \rvert \le \sqrt{5}$	✓ ½
4	So x lies in either of two intervals; hence the domain is: $x \in \left[-\sqrt{5}, -\sqrt{3}\right] \cup \left[\sqrt{3}, \sqrt{5}\right]$	✓ ½

Answer $\left[-\sqrt{5}, -\sqrt{3}\right] \cup \left[\sqrt{3}, \sqrt{5}\right]$

Watch out: Writing only $\left[\sqrt{3}, \sqrt{5}\right]$ and forgetting the negative interval.

10 **Short answer** 2 marks

Find the domain of $f(x) = \sin^{-1} x + \cos^{-1}(x - 1)$.

Formula / idea: Domain of a sum is the intersection of the domains of its terms.

1	$\sin^{-1} x$ is defined when: $-1 \le x \le 1 \;\Rightarrow\; x \in [-1, 1]$	✓ ½
2	$\cos^{-1}(x - 1)$ needs $-1 \le x - 1 \le 1$; add 1 to each part. $0 \le x \le 2 \;\Rightarrow\; x \in [0, 2]$	✓ ½
3	Both terms must be defined, so take the intersection. $[-1, 1] \cap [0, 2]$	✓ ½
4	Hence the domain of f is: $[0, 1]$	✓ ½

Answer $[0, 1]$

Watch out: Taking the union $[-1, 2]$ instead of the intersection.

11 **Short answer** 2 marks

Find the value of $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)+\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$.

Formula / idea: $\cos^{-1}(\cos x)=x$ for $x\in[0,\pi]$; $\tan^{-1}(\tan x)=x$ for $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

1 $\frac{13\pi}{6}\notin[0,\pi]$; use $\cos(2\pi+\theta)=\cos\theta$.
$$\cos\frac{13\pi}{6}=\cos\left(2\pi+\frac{\pi}{6}\right)=\cos\frac{\pi}{6}$$ ✓ ½

2 Since $\frac{\pi}{6}\in[0,\pi]$:
$$\cos^{-1}\left(\cos\frac{13\pi}{6}\right)=\cos^{-1}\left(\cos\frac{\pi}{6}\right)=\frac{\pi}{6}$$ ✓ ½

3 $\frac{7\pi}{6}\notin\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$; use $\tan(\pi+\theta)=\tan\theta$.
$$\tan^{-1}\left(\tan\frac{7\pi}{6}\right)=\tan^{-1}\left(\tan\frac{\pi}{6}\right)=\frac{\pi}{6}$$ ✓ ½

4 Add the two values.
$$\frac{\pi}{6}+\frac{\pi}{6}=\frac{\pi}{3}$$ ✓ ½

Answer $\frac{\pi}{3}$

Watch out: Writing $\frac{13\pi}{6}+\frac{7\pi}{6}$: both angles are outside the principal branches.

12 **Short answer** 2 marks

Evaluate: $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]$.

Formula / idea: $\sin^{-1}(-x)=-\sin^{-1}x$

1 Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$.
$$\sin^{-1}\left(-\frac{1}{2}\right)=-\sin^{-1}\left(\frac{1}{2}\right)=-\frac{\pi}{6}$$ ✓ 1

2 Substitute in the bracket.
$$\frac{\pi}{3}-\left(-\frac{\pi}{6}\right)=\frac{\pi}{3}+\frac{\pi}{6}=\frac{\pi}{2}$$ ✓ ½

3 Therefore:
$$\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]=\sin\frac{\pi}{2}=1$$ ✓ ½

Answer 1

Watch out: Sign slip: $\frac{\pi}{3}-\left(-\frac{\pi}{6}\right)$ is $\frac{\pi}{2}$, not $\frac{\pi}{6}$.

13 **Short answer** 3 marks

Find the value of $\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$.

Formula / idea: Work from the innermost bracket outwards.

1 Innermost value: $\sin\frac{\pi}{6}=\frac{1}{2}$ and $\frac{\pi}{6}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. ✓ 1
$$\sin^{-1}\frac{1}{2}=\frac{\pi}{6}$$

2 Double it.
$$2\sin^{-1}\frac{1}{2}=2\times\frac{\pi}{6}=\frac{\pi}{3}$$

3 Find the cosine and multiply by 2. ✓ 1
$$2\cos\frac{\pi}{3}=2\times\frac{1}{2}=1$$

4 So the expression becomes $\tan^{-1}(1)$.
$$\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]=\tan^{-1}(1)$$

5 Since $\tan\frac{\pi}{4}=1$ and $\frac{\pi}{4}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$: ✓ 1
$$\tan^{-1}(1)=\frac{\pi}{4}$$

Answer $\frac{\pi}{4}$

Watch out: Doubling inside the sine: $2\sin^{-1}\frac{1}{2}$ is $\frac{\pi}{3}$, not $\sin^{-1}(1)$.

14 **Short answer** 3 marks

Evaluate: $\sec^2\left(\tan^{-1}2\right)+\csc^2\left(\cot^{-1}3\right)$.

Formula / idea: $\sec^2\theta=1+\tan^2\theta$, $\csc^2\theta=1+\cot^2\theta$; $\tan\left(\tan^{-1}x\right)=x$ for all real x.

1 Use $\sec^2\theta=1+\tan^2\theta$ with $\theta=\tan^{-1}2$. ✓ 1
$$\sec^2\left(\tan^{-1}2\right)=1+\left[\tan\left(\tan^{-1}2\right)\right]^2$$

2 Since $\tan\left(\tan^{-1}2\right)=2$: ✓ ½
$$=1+2^2=5$$

3 Use $\csc^2\theta=1+\cot^2\theta$ with $\theta=\cot^{-1}3$. ✓ 1
$$\csc^2\left(\cot^{-1}3\right)=1+\left[\cot\left(\cot^{-1}3\right)\right]^2=1+3^2=10$$

4 Add the two results.
$\sec^2\left(\tan^{-1} 2\right) + \csc^2\left(\cot^{-1} 3\right) = 5 + 10 = 15$ ✓ ½

Answer 15

Watch out: Writing $\sec^2\left(\tan^{-1} 2\right) = 2^2 = 4$, forgetting the $1+$ in the identity.

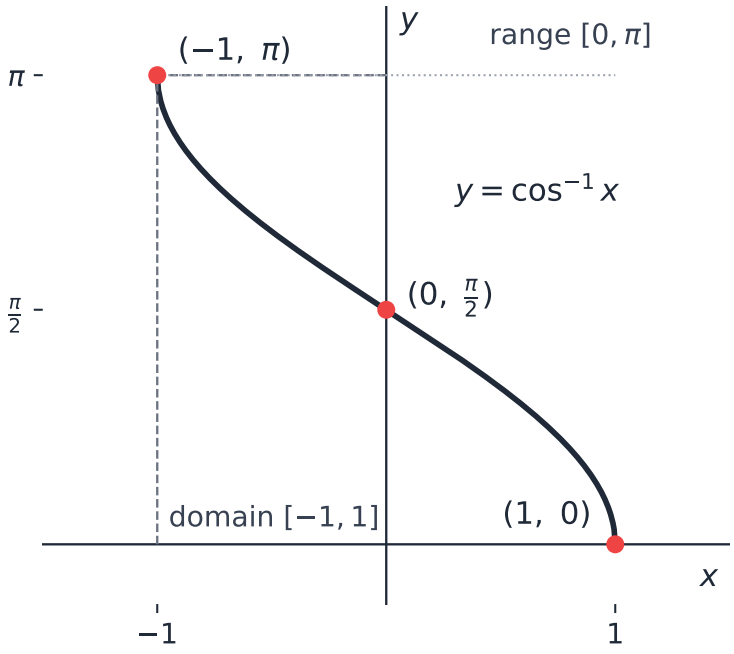
Level 3

19 marks

15 **Long answer** 3 marks

Draw the graph of $y = \cos^{-1} x$ for its principal value branch. Using the graph, write its domain and range, and state whether the function is increasing or decreasing.

Formula / idea: Graph of $\cos^{-1} x$ is the mirror image of $y = \cos x$, $x \in [0, \pi]$, in the line $y = x$.



1 Make a table of values using the principal branch $[0, \pi]$. ✓ ½

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
$y = \cos^{-1} x$	π	$\frac{2\pi}{3}$	$\frac{\pi}{2}$	$\frac{\pi}{3}$	0

2 Plot these points and join them by a smooth curve (see graph). ✓ 1
Curve runs from $(-1, \pi)$ through $\left(0, \frac{\pi}{2}\right)$ to $(1, 0)$

3 The curve spreads from $x = -1$ to $x = 1$, so: ✓ ½
$\text{Domain} = [-1, 1]$

4 The curve goes from height 0 up to height π, so: ✓ ½
$\text{Range} = [0, \pi]$

5 As x increases from -1 to 1, y falls from π to 0.
$-1 < 0$ but $\cos^{-1}(-1) = \pi > \cos^{-1}(0) = \dfrac{\pi}{2}$

6 Hence $y = \cos^{-1} x$ is a **decreasing** function on $[-1, 1]$. ✓ ½

Answer **Domain $[-1, 1]$; range $[0, \pi]$; $y = \cos^{-1} x$ is a decreasing function.**

Watch out: Drawing the curve rising from left to right; $\cos^{-1} x$ falls, unlike $\sin^{-1} x$.

16 **Long answer** 3 marks

Find the value of $\sin^{-1}\left(\sin \frac{4\pi}{5}\right) + \cos^{-1}\left(\cos \frac{4\pi}{3}\right) + \tan^{-1}\left(\tan \frac{3\pi}{4}\right)$.

Formula / idea: Bring each angle into its principal branch: $\sin^{-1}\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $\cos^{-1}[0, \pi]$, $\tan^{-1}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

1 $\frac{4\pi}{5} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; use $\sin(\pi - \theta) = \sin\theta$. ✓ ½
$\sin\dfrac{4\pi}{5} = \sin\left(\pi - \dfrac{4\pi}{5}\right) = \sin\dfrac{\pi}{5}$

2 Since $\frac{\pi}{5} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$: ✓ ½
$\sin^{-1}\left(\sin\dfrac{4\pi}{5}\right) = \dfrac{\pi}{5}$

3 $\frac{4\pi}{3} \notin [0, \pi]$; use $\cos(2\pi - \theta) = \cos\theta$. ✓ ½
$\cos\dfrac{4\pi}{3} = \cos\left(2\pi - \dfrac{4\pi}{3}\right) = \cos\dfrac{2\pi}{3}$

4 Since $\frac{2\pi}{3} \in [0, \pi]$: ✓ ½
$\cos^{-1}\left(\cos\dfrac{4\pi}{3}\right) = \dfrac{2\pi}{3}$

5 $\frac{3\pi}{4} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; use $\tan(\theta - \pi) = \tan\theta$. ✓ ½
$\tan^{-1}\left(\tan\dfrac{3\pi}{4}\right) = \tan^{-1}\left(\tan\left(-\dfrac{\pi}{4}\right)\right) = -\dfrac{\pi}{4}$

6 Add the three values (LCM of $5, 3, 4$ is 60). ✓ ½
$\dfrac{\pi}{5} + \dfrac{2\pi}{3} - \dfrac{\pi}{4} = \dfrac{12\pi + 40\pi - 15\pi}{60} = \dfrac{37\pi}{60}$

Answer $\frac{37\pi}{60}$

Watch out: Using $\cos(\pi + \theta)$ for $\frac{4\pi}{3}$: that changes the sign, so $\frac{\pi}{3}$ is wrong.

17 **Long answer** 3 marks

Find the domain and range of $f(x) = 2\sin^{-1}(1-x)$.

Formula / idea: $\sin^{-1} t$ needs $-1 \le t \le 1$ and gives values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

1	**Domain:** the input of $\sin^{-1}$ must lie in $[-1, 1]$. $-1 \le 1 - x \le 1$	✓ ½
2	Subtract 1 from each part. $-2 \le -x \le 0$	✓ ½
3	Multiply by -1; the inequality signs reverse. $0 \le x \le 2 \;\Rightarrow\; \text{Domain} = [0, 2]$	✓ ½
4	**Range:** principal values of $\sin^{-1}$ lie in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. $-\frac{\pi}{2} \le \sin^{-1}(1-x) \le \frac{\pi}{2}$	✓ ½
5	Multiply each part by 2. $-\pi \le 2\sin^{-1}(1-x) \le \pi$	✓ ½
6	Hence the range of f is: $[-\pi, \pi]$	✓ ½

Answer **Domain** $[0, 2]$**; range** $[-\pi, \pi]$

Watch out: Forgetting to reverse the signs when multiplying by -1, giving $[-2, 0]$.

18 **Long answer** 3 marks

Write $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ in the simplest form.

Formula / idea: Substitute $x = \tan\theta$ so that $1 + x^2 = \sec^2\theta$.

1	Put $x = \tan\theta$. $x = \tan\theta \;\Rightarrow\; \theta = \tan^{-1} x, \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	✓ ½
2	Simplify the root using $1 + \tan^2\theta = \sec^2\theta$. $\sqrt{1+x^2} = \sqrt{1+\tan^2\theta} = \sqrt{\sec^2\theta} = \sec\theta$ $\sec\theta > 0$ for $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	✓ 1
3	Simplify the fraction. $\frac{x}{\sqrt{1+x^2}} = \frac{\tan\theta}{\sec\theta} = \frac{\sin\theta}{\cos\theta} \times \cos\theta = \sin\theta$	✓ ½

4 Since $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, which is inside the branch of $\sin^{-1}$:

$$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \sin^{-1}(\sin\theta) = \theta$$

✓ ½

5 Substitute back $\theta = \tan^{-1} x$.

$$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \tan^{-1} x$$

✓ ½

Answer $\tan^{-1} x$

Watch out: Leaving the answer as θ; always substitute back $\theta = \tan^{-1} x$.

19 **Long answer** 3 marks

Write $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$, $0 < x < \pi$, in the simplest form.

Formula / idea: $1 - \cos x = 2\sin^2\frac{x}{2}$ and $1 + \cos x = 2\cos^2\frac{x}{2}$.

1 Use the half-angle identities.

$$\frac{1-\cos x}{1+\cos x} = \frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}}$$

✓ 1

2 Cancel 2 and write as tangent.

$$\frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}} = \tan^2\frac{x}{2}$$

✓ ½

3 Since $0 < x < \pi$, we have $0 < \frac{x}{2} < \frac{\pi}{2}$, so $\tan\frac{x}{2} > 0$.

4 Take the square root.

$$\sqrt{\tan^2\frac{x}{2}} = \tan\frac{x}{2}$$

✓ ½

5 So the expression becomes:

$$\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}} = \tan^{-1}\left(\tan\frac{x}{2}\right)$$

6 Since $\frac{x}{2} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, the principal branch of $\tan^{-1}$:

$$\tan^{-1}\left(\tan\frac{x}{2}\right) = \frac{x}{2}$$

✓ 1

Answer $\frac{x}{2}$

Watch out: Skipping the sign check: $\sqrt{\tan^2\frac{x}{2}} = \left|\tan\frac{x}{2}\right|$, positive here only because $0 < \frac{x}{2} < \frac{\pi}{2}$.

20 **Case study** 4 marks

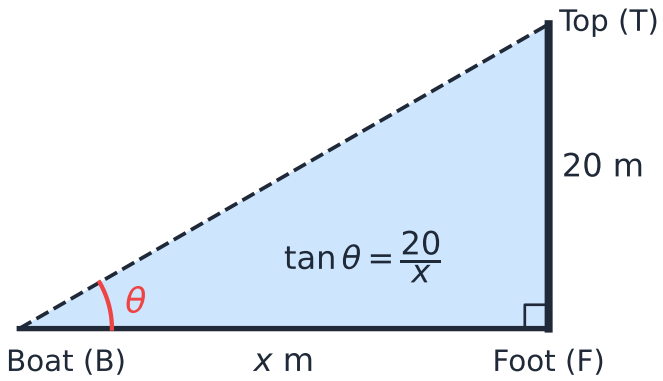
Case Study: A lighthouse is 20 m tall. A boat is at a horizontal distance of x m from the foot of the lighthouse. The angle of elevation θ of the top of the lighthouse from the boat is given by $\theta = \tan^{-1}\left(\frac{20}{x}\right)$.

(i) Find θ when $x = 20\sqrt{3}$ m. 1 mark

(ii) Write the domain and range of the principal value branch of $\tan^{-1} x$. 1 mark

(iii) Find the distance x of the boat when $\theta = \frac{\pi}{3}$. Also find the principal value of $\tan^{-1}\left(-\sqrt{3}\right)$. 2 marks

Formula / idea: In the right triangle, $\tan\theta = \frac{\text{height}}{\text{distance}} = \frac{20}{x}$.



(i) 1 mark

1 Put $x = 20\sqrt{3}$ in $\theta = \tan^{-1}\left(\frac{20}{x}\right)$. ✓ ½

$$\theta = \tan^{-1}\left(\frac{20}{20\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

2 Since $\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$ and $\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: ✓ ½

$$\theta = \frac{\pi}{6}$$

Answer (i) $\theta = \frac{\pi}{6}$

(ii) 1 mark

1 $\tan^{-1} x$ is defined for every real number. ✓ ½

$$\text{Domain} = \mathbb{R}$$

2 Its principal value branch is the open interval: ✓ ½

$$\text{Range} = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$\tan\theta$ is not defined at $\pm\frac{\pi}{2}$

Answer (ii) **Domain $\mathbb{R}$; range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$**

(iii) 2 marks

1	$\theta = \tan^{-1}\left(\frac{20}{x}\right)$ gives $\tan\theta = \frac{20}{x}$; put $\theta = \frac{\pi}{3}$. $\tan\frac{\pi}{3} = \frac{20}{x} \Rightarrow \sqrt{3} = \frac{20}{x}$	✓ ½
2	Solve for x and rationalise. $x = \frac{20}{\sqrt{3}} = \frac{20\sqrt{3}}{3}\ \text{m} \approx 11.55\ \text{m}$	✓ ½
3	Use $\tan^{-1}(-x) = -\tan^{-1}x$. $\tan^{-1}\left(-\sqrt{3}\right) = -\tan^{-1}\left(\sqrt{3}\right)$	✓ ½
4	Since $\tan\frac{\pi}{3} = \sqrt{3}$ and $-\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: $\tan^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{3}$	✓ ½

Answer (iii) $x = \frac{20\sqrt{3}}{3}$ **m** ≈ 11.55 **m;** $\tan^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{3}$

Watch out: Writing $\tan^{-1}\left(-\sqrt{3}\right) = \frac{2\pi}{3}$; it is outside $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.